Experimental Physiology – *Research Paper*

Effects of exercise on muscle glycogen synthesis signalling and enzyme activities in pigs carrying the *PRKAG3* mutation

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The dominant RN$^-$ mutation in pigs results in excessive glycogen storage in skeletal muscle. The mutation is situated in the *PRKAG3* gene, which encodes a muscle-specific isoform of the AMP-activated protein kinase (AMPK) $\gamma 3$ subunit. AMPK is an important regulator of carbohydrate and fat metabolism in mammalian cells. The aim of the present study was to examine the effect of exercise on glycogen synthesis signalling pathways in muscle and to study enzyme activities of importance in carbohydrate metabolism in pigs with or without the *PRKAG3* mutation. Glycogen content, metabolic enzyme activities and expression or phosphorylation of signalling proteins were analysed in skeletal muscle specimens obtained at rest, after a single treadmill exercise bout and after 3 h recovery. The *PRKAG3* mutation carriers had higher glycogen content, a tendency for lower expression of AMPK ($P < 0.07$) and higher hexokinase and phosphorylase activities, whereas citrate synthase, 3-hydroxyacyl-CoA dehydrogenase and glycogen synthase activities did not differ between genotypes. Carriers and non-carriers of the RN$^-$ mutation showed a similar degradation of glycogen after exercise, whereas the rate of resynthesis was faster in the carriers. Acute exercise stimulated Akt phosphorylation on Ser^{473} in both genotypes, and the effect was greater in the carriers than in the non-carriers. Acute exercise also stimulated phosphorylation of Akt substrate of 160 kDA and Glycogen synthase kinase 3 in the carriers and GSK3α in the non-carriers. In conclusion, the increased rate of glycogen synthesis following exercise in pigs carrying the *PRKAG3* mutation correlates with an increased signalling response of Akt and its substrate, AS160, and a higher activity of hexokinase, indicating an increased glucose influx and phosphorylation of glucose, directed towards glycogen synthesis.

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The naturally occurring dominant RN$^-$ (Rendement Napole) mutation in Hampshire pigs results in an excessive amount of glycogen storage in skeletal muscle, especially of macroglycogen, whereas the glycogen concentration in liver is unchanged (Monin *et al.* 1992; Enfält *et al.* 1997; Essén-Gustavsson *et al.* 2005). The marked increase in glycogen content is primarily observed in white muscle of pigs (Estrade *et al.* 1993). These white muscles are shown to be more oxidative, since the activities of citrate synthase (CS) and 3-hydroxyacyl-CoA dehydrogenase (HAD) are increased and the activity of lactate dehydrogenase (LDH) decreased (Estrade *et al.* 1994; Lebret *et al.* 1999).

The first step in the pathway of glycogen synthesis is the transport of glucose across the muscle membrane. This is facilitated by glucose transporter proteins, of which Glucose transporter 4 (GLUT4) is the most

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important. When glucose has entered the muscle cell, it is phosphorylated by the enzyme hexokinase (HK) to glucose-6-phoshate (G-6-P). This metabolite can then either be converted in the glycogen synthesis pathway to glycogen or be converted in the glycolysis pathway to pyruvate. Pyruvate is the end-product of glycolysis, which can either be converted to acetyl-CoA for oxidation in the citric acid cycle or be converted anaerobically to lactate via the enzyme LDH. Important enzymes in the pathway of glycogen synthesis are UDP-glucose pyrophosphorylase, glycogen synthase (GS) and glycogen branching enzyme. The branching enzyme activity was twice as high in the muscles of the carriers (Estrade *et al.* 1994), and the expression and activity of UDP-glucose pyrophosphorylase was upregulated in pigs carrying the RN^- mutation (Hedegaard *et al.* 2004). Enzymes involved in glycogenolysis are phosphorylase and the glycogen debranching enzyme, and these do not differ between carriers and non-carriers of the mutation (Estrade *et al.* 1994).

This RN^- mutation is a substitution in the *PRKAG3* gene, which encodes a muscle-specific isoform of the AMP-activated protein kinase (AMPK) $\gamma 3$ subunit (Milan *et al.* 2000). AMPK is a multifactorial metabolic sensor that responds to shifts in cellular energy levels. It is known to inhibit enzymes involved in the synthesis of glycogen and to stimulate glucose uptake (Merrill *et al.* 1997; Ojuka *et al.* 2000). AMPK has also been shown to be an exercise-responsive signalling intermediate to glucose transport in skeletal muscle (Mu *et al.* 2001; Wright *et al.* 2004). The function of AMPK is closely related to glycogen storage, and it phosphorylates glycogen synthase *in vitro* (Carling & Hardie, 1989) and co-immunoprecipitates with glycogen synthase and glycogen phosphorylase from skeletal muscle (Chen *et al.* 1999). Since AMPK is a key metabolic enzyme, the *PRKAG3* mutation is expected to affect the regulation of several key metabolic processes in the cell. The activity of AMPK has been shown to be decreased in pigs with the *PRKAG3* mutation (Milan *et al.* 2000).

Akt/protein kinase B is an important kinase downstream of the phosphatidyl inositol 3-kinase cascade in the insulin signalling pathway (Vanhaesebroeck & Alessi, 2000), and is important for stimulating glycogen synthesis and promoting glucose storage in muscle (Ueki *et al.* 1998). Akt inhibits glycogen synthase kinase 3 (GSK3) via direct phosphorylation (Cross *et al.* 1995) and decreases its inhibitory action towards glycogen synthase, which leads to an increase in the rate of glycogen synthesis. Two isoforms of GSK3, α and β, have been identified (Woodgett, 1990, 1991), and exercise has been shown to regulate Akt and GSK3α in human skeletal muscle (Sakamoto *et al.* 2003).

An Akt substrate of 160 kDa, AS160, is the most proximal step identified in the insulin signalling transduction pathway (Kane *et al.* 2002; Sano *et al.* 2003). It plays an important role by stimulating GLUT4 translocation, resulting in movement of more glucose into the cell, which is of importance for glycogen synthesis. Increased AS160 phosphorylation has also been observed in rats in response to contractile activity (Bruss *et al.* 2005) and in humans after exercise (Deshmukh *et al.* 2006).

Pigs carrying the *PRKAG3* mutation use glycogen during treadmill exercise to the same extent as non-carriers, but the rate of glycogen resynthesis after exercise is faster in the carriers (Essén-Gustavsson *et al.* 2005). The hypothesis is that enzyme activities involved in the glycogen synthesis pathway and the signalling response to exercise, involving AMPK and Akt in skeletal muscle, would differ between genotypes and benefit glycogen synthesis in pigs carrying the *PRKAG3* mutation.

The aim of this study was to investigate the effect of exercise on glycogen synthesis signalling in pigs that are carriers and non-carriers of the *PRKAG3* mutation and to further study enzyme activities involved in carbohydrate metabolism.

Methods

Animals

The experiments were approved by the Ethical Committee for Animal Experiments, Uppsala, Sweden.

Sixteen clinically healthy female pigs (Yorkshire/Swedish Landrace × Hampshire) from the University herd were used. The pigs were 9–11 weeks old, had a mean weight of 29 ± 2.5 kg when they arrived, and were stalled at the department. Eight pigs were carriers and eight pigs were non-carriers of the *PRKAG3* mutation as verified by DNA analyses of blood (Milan *et al.* 2000). The pigs were housed in pens (4 pigs per pen) with a concrete floor, with straw as bedding. The animals were fed a commercial finisher diet, Piggfor®; Origio 522 PK, (Lantmännen, Uppsala, Sweden) *ad libitum* and had free access to water. The pigs were inspected daily for their clinical state of health.

Experimental design

The study was conducted during a 4 week period and started with a 2 week period of acclimation of the animals. The pigs were also given time to get accustomed to running on the treadmill (Säto, Knivsta, Sweden) and were allowed to walk and trot on it for a few minutes on four separate days before an exercise test was conducted. Following acclimation, the pigs were anaesthetized and muscle biopsy specimens obtained from the middle part of the right biceps femoris muscle at a depth of 3–4 cm using a needle-biopsy technique (Bergström, 1962). The muscle specimens were immediately frozen in liquid nitrogen and stored at −80°C until analysed. Anaesthesia was achieved

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by administration of a combination of medetomidine (Domitor® vet, 1 mg ml^{-1}; Orion Pharma AB, Stockholm, Sweden) and a proprietary mixture of tiletamine and zolazepam (Zoletil®, 250 mg ml^{-1} of tiletamine and 250 mg ml^{-1} of zolazepam; Reading, Carros, France). The mixture contained 1 mg ml^{-1} of medetomidine and 100 mg ml^{-1} of tiletamine–zolazepam and was given i.m. at the dosage of 0.05 ml (kg body weight)$^{-1}$.

Three days after the resting biopsies had been obtained, all pigs ran 700 m on a treadmill at a speed of 1.5 m s^{-1}. Within 10 min after completion of the exercise, local anaesthetic (mepivacaine; Carbocain® adrenaline, 20 mg ml^{-1} + 5 μg ml^{-1}; Astra Zeneca AB, Södertälje, Sweden) was administered locally and a biopsy specimen obtained from the left biceps femoris muscle. The pigs were restrained at the front of the treadmill using a board to enable biopsy specimens to be collected. After 3 h of recovery from the exercise, a biopsy from right *m. biceps femoris* was collected using the same anaesthesic technique.

Muscle analyses

Muscle biopsies were freeze-dried overnight and then the muscle tissue was dissected out under a microscope to remove visible blood, fat and connective tissue. Samples were subsequently homogenized in ice-cold homogenization buffer containing 20 mM Tris (pH 7.8), 137 mM NaCl, 2.7 mM KCl, 1 mM $MgCl_2$, 1% Triton X-100, 10% (w/v) glycerol, 10 mM NaF, 1 mM EDTA, 5 mM sodium pyrophosphate, 0.5 mM Na_3V0_4, 1 μg ml^{-1} leupeptin, 0.2 mM phenylmethylsulphonyl fluoride, 1 μg ml^{-1} aprotinin, 1 mM dithiothreitol, 1 mM benzamidine and 1 μM microcystin. The homogenized samples were then rotated for 30 min at 4°C and thereafter subjected to centrifugation (12 000*g* for 15 min at 4°C). Protein concentration was determined in the supernatant using the BCA protein assay kit (Pierce, Rockford, IL, USA). Phosphorylation of protein kinases was determined by immunoblot analysis. Skeletal muscle protein lysate (40 μg) was separated by SDS-PAGE (7.5% resolving gel), transferred to polyvinylidene flouride membranes and blocked with 7.5% non-fat milk in tris buffered saline Tween-20 (TBS-T) for 2 h at room temperature. Membranes were incubated overnight at 4°C on a shaking platform with the following primary antibodies: anti-pGSK3α/β Ser$^{21/9}$, pAS160, AS160, Akt, pAkt Thr^{308}, pAkt Ser^{473}, panAMPKα, pAMPK Thr^{172} (Cell Signaling Technology, Boston, MA, USA), pAcetyl-CoA carboxylase (ACC) Ser^{79} (Upstate/Millipore Corporate Headquarters, Billerica, MA, USA) or GS (gift from Professor Erik Richter, Copenhagen University). Thereafter, membranes were washed with TBS-T and incubated with appropriate anti-rabbit or anti-mouse secondary antibody. Samples from pigs with or without the *PRKAG3* mutation were included and analysed on the same gel in parallel. Proteins were visualized by enhanced chemiluminescence (ECL, Amersham, Little Chalfont, Buckinghamshire, UK) and quantified by densitometry and Molecular Analyst Software (Bio-Rad, Richmond, CA, USA). All membranes were normalized for loading with glyceraldehyde-3-phophate dehydrogenase (GAPDH; Santa Cruz Biotechnology, Santa Cruz, CA, USA).

For glycogen determination, 1–2 mg of pure tissue was boiled in 1 M HCl for 2 h to form glucose residues. Glucose was analysed by a fluorometric method (Lowry & Passoneau, 1973).

To determine the activities of CS, HAD, LDH, HK and phosphorylase, 1–2 mg of pure tissue was homogenized with an ultrasound disintegrator (Branson, Lutterworth, Leicestershire, UK) in ice-chilled potassium phosphate buffer (0.1 M, pH 7.3) at a dilution of 1:400 and then analysed by fluorometry (Lowry & Passoneau, 1973).

Glycogen synthase activity was determined by a method modified from Thomas *et al.* (1968), based on incorporation of ^{14}C-labelled uridine diphosphate-glucose into glycogen. The active form of GS activity (that is the form of Glycogen synthase which is activated by insulin, low levels of glycogen and muscle contraction) was measured at a low concentration of glucose-6-phosphate (0.3 mM) and the total activity at a high concentration (6.0 mM).

Statistics

Data are presented as means ± S.E.M. in the figures. A rank sum test was used for comparison of values between the carrier and the non-carrier pigs. For comparison of the two muscle biopsy specimens within carriers or non-carriers, a signed rank test was used. The level of statistical significance was set at $P < 0.05$. Statistical analyses were carried out using SigmaStat statistical software version 11.0 (Aspire Software International, 20448 Charter Oak Drive, Ashburn, VA, 20147, USA).

Results

Resting conditions

As expected, the carriers of the *PRKAG3* mutation had higher glycogen content than the non-carriers before exercise (Fig. 1*A*).

The enzyme activities of HK and phosphorylase were higher in carriers, but no differences were observed concerning CS, HAD and LDH or in the percentage of CS, HAD, LHD and GS were observed relative to the total (Tables 1 and 2).

There were no differences in pAMPK Thr^{172} (62 kDa), pACC^{Ser79} (280 kDa), pAkt Ser^{473} (60 kDa),

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Table 1. Enzyme activities in biceps femoris muscle at rest, immediately after exercise and after 3 h of recovery

	CS (mmol kg^{-1} min^{-1})		HK (mmol kg^{-1} min^{-1})		LDH (mmol kg^{-1} min^{-1})		Phosphorylase (mmol kg^{-1} min^{-1})		HAD (mmol kg^{-1} min^{-1})	
	Non-carriers	Carriers	Non-carriers	Carriers	Non-carriers	Carriers	Non-carriers	Carriers	Non-carriers	Carriers
Rest	22.6 ± 0.6	21 ± 0.6	4.4 ± 0.2	8.0 ± 0.4*	1919 ± 108	1520 ± 176	15.5 ± 0.4	18.5 ± 0.7*	26.0 ± 0.2	25.4 ± 0.3
Postxercise	21.3 ± 1.5	22.4 ± 1.0	5.1 ± 0.2	7.9 ± 0.3*	2072 ± 182	1572 ± 89*	17.0 ± 0.4	20.3 ± 0.5*	26.1 ± 0.9	25.3 ± 0.5
Recovery	19.8 ± 1	21.2 ± 0.5	4.0 ± 0.4	8.2 ± 0.5*	1700 ± 86†	1525 ± 151	15.3 ± 1.4	20.5 ± 0.6*	26.1 ± 0.9	24.7 ± 0.8

Data are expressed as means ± S.E.M. Abbreviations: CS, citrate synthase; HAD, 3-hydroxyacyl-CoA dehydrogenase; HK, hexokinase; and LDH, lactate dehydrogenase. * Significantly different ($P < 0.05$) in carriers *versus* non-carriers. † Significantly different ($P < 0.05$) from postexercise.

Table 2. Enzyme activities in biceps femoris muscle at rest, immediately after exercise and after 3 h of recovery

	Glycogen synthase (total activity (nmol mg^{-1} min^{-1}))		Glycogen synthase (% of active form from total)	
	Non-carriers	Carriers	Non-carriers	Carriers
Rest	139 ± 11	151 ± 10	28.6 ± 0.1	29.6 ± 0.1
Postexercise	139 ± 15	158 ± 6	32 ± 0.1	29.2 ± 0.1*
Recovery	125 ± 6	160 ± 10	33.6 ± 0.1	29.2 ± 0.1

Data are expressed as means ± S.E.M. * Significantly different ($P < 0.05$) in carriers *versus* non-carriers. † Significantly different ($P < 0.05$) from postexercise.

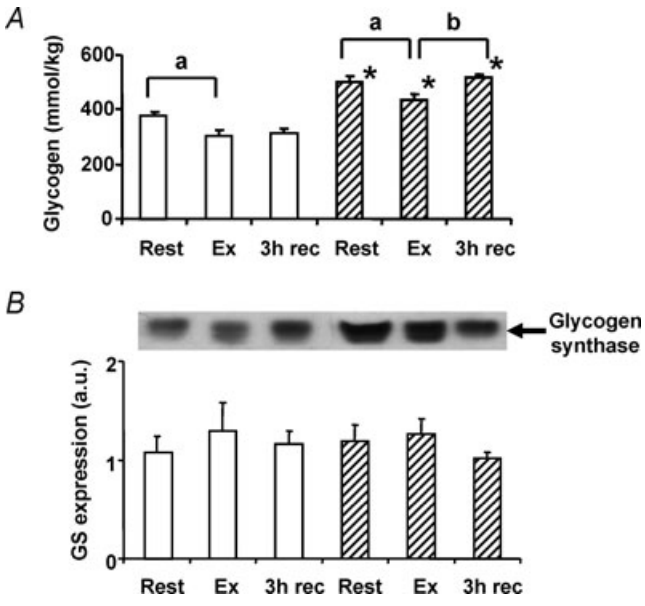


Figure 1. Effect of acute exercise on glycogen content and glycogen synthase (GS) expression in skeletal muscle from non-carriers (open bars, $n = 8$) and carriers of the *PRKAG3* mutation (hatched bars, $n = 8$)
Mean values of glycogen content (*A*) and expression of GS (*B*) are shown from biopsies taken from biceps femoris at rest (Rest), immediately after exercise (Ex; 700 m, 1.5 m s^{-1}) and after 3 h of recovery (3 h rec). A representative Western blot image of GS is shown in *B*. *$P < 0.05$ significantly different in carriers *versus* non-carriers; [a]$P < 0.05$ significantly different from rest; and [b]$P < 0.05$ significantly different from postexercise.

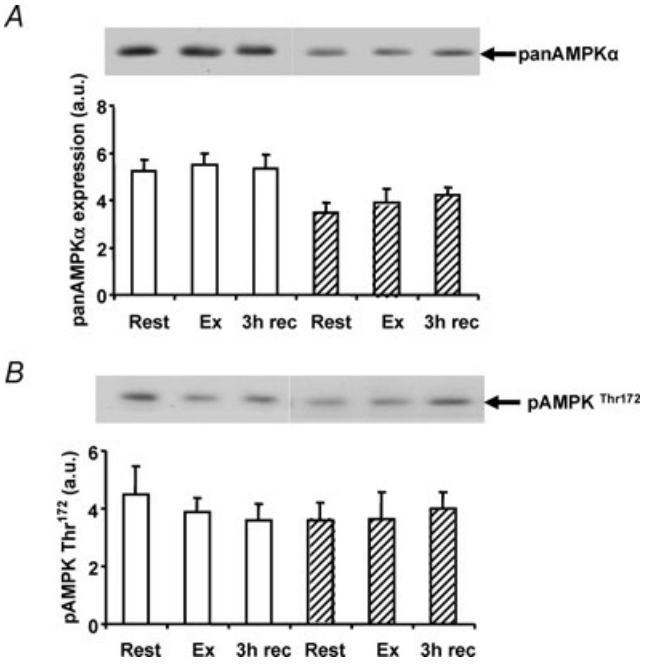


Figure 2. Expression and phosphorylation of AMPK in skeletal muscle from non-carriers (open bars, $n = 8$) and carriers of the *PRKAG3* mutation (hatched bars, $n = 7$) in response to acute exercise
Mean values of expression (*A*) and phosphorylation (*B*) of AMPK are shown from biopsies taken from biceps femoris at rest (rest), immediately after exercise (Ex; 700 m, 1.5 m s^{-1}) and after 3 h of recovery (3 h rec). Representative Western blot images are also shown.

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pAS160 (160 kDa), pGSK3α (51 kDa), pGSK3β (46 kDa) and expression of GS (90 kDa) between genotypes (Figs 1–5). Expression of Akt (60 kDa) was higher in carriers, and the expression of AMPK (62 kDA) showed a tendency to be lower in the carriers ($P < 0.07$; Figs 2*A* and 4*A*). The ratio of phosphorylated AMPK to total AMPK (pAMPK/tAMPK) did not differ between the genotypes. Furthermore, no changes in expression of AS160 (160 kDa) and phosphorylation of Akt Thr^{308} (60 kDA) were observed (data not shown).

Acute exercise and recovery

The glycogen content decreased after exercise in both carriers (by 64 mmol kg^{-1}) and non-carriers (by 75 mmol kg^{-1}), with no difference between genotypes (Fig. 1*A*). After 3 h of recovery, the *PRKAG3* mutation carriers had increased muscle glycogen content (by 81 mmol kg^{-1}), while no such elevation was observed in the non-carriers.

The activities of HK and phosphorylase were higher in the carriers than in the non-carriers after exercise and 3 h into the recovery phase. After exercise, the activity of LDH and the activity of LDH and GS showed a slight decrease in the carriers. Levels returned to baseline after 3 h of recovery (Tables 1 and 2).

The carriers had a tendency to a lower expression of AMPK (Fig. 2*A*). The phosphorylation of AMPK and the pAMPK/tAMPK ratio did not differ statistically between genotypes after exercise or after 3 h of recovery (Fig. 2*B*). Acute exercise led to an increase in pAkt Ser^{473} (Fig. 4*B*) in both genotypes, and the increase in pAkt Ser^{473} was higher in the carriers (6.3-fold) than in the non-carriers (3.1-fold). After 3 h of recovery, phosphorylation of Akt at Ser^{473} decreased in both genotypes, although the carriers retained a higher (2.2-fold) phosphorylation of Akt than the non-carriers. The carriers had a higher phosphorylation of AS160 (1.8-fold) after exercise than at rest (Fig. 3*B*). A higher phosphorylation of GSK3α was seen after exercise in both carriers (1.9-fold) and non-carriers (1.4-fold; Fig. 5*A*). Exercise and recovery did

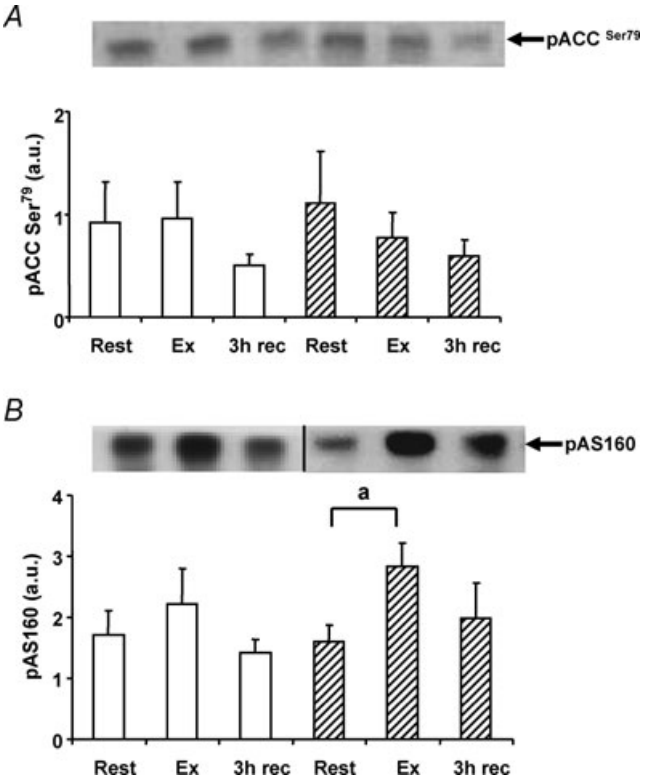


Figure 3. Phosphorylation of Acetyl-CoA carboxylase (ACC) and Akt substrate of 160 kDA (AS160) in skeletal muscle from non-carriers (open bars, $n = 8$) and carriers of the *PRKAG3* mutation (hatched bars, $n = 7$) in response to acute exercise
Mean values of phosphorylation of ACC (*A*) and AS160 (*B*) are shown from biopsies taken from biceps femoris at rest (Rest), immediately after exercise (Ex; 700 m, 1.5 m s^{-1}) and after 3 h of recovery (3 h rec). Representative Western blot images are also shown. $^{a}P < 0.05$ significantly different from rest.

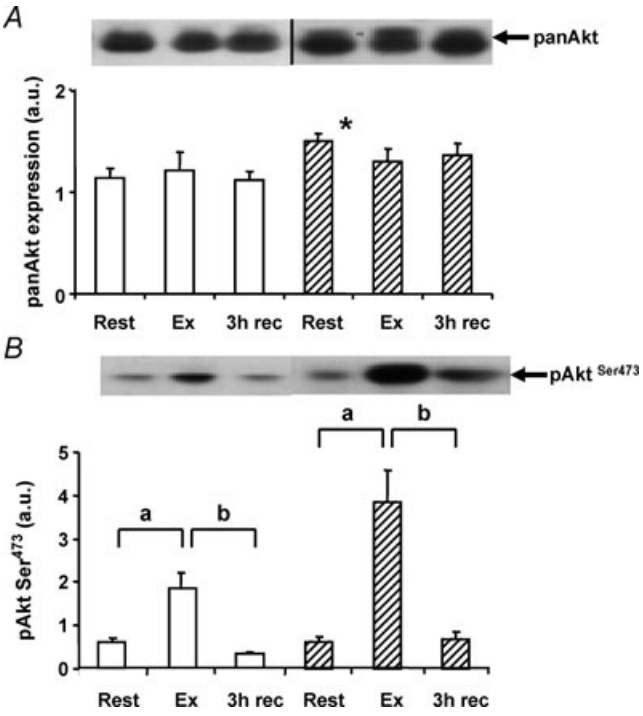


Figure 4. Expression and phosphorylation of Akt in skeletal muscle from non-carriers (open bars, $n = 8$) and carriers of the *PRKAG3* mutation (hatched bars, $n = 8$) in response to acute exercise
Mean values of expression (*A*) and phosphorylation of Akt (*B*) are shown from biopsies taken from biceps femoris at rest (rest), immediately after exercise (Ex; 700 m, 1.5 m s^{-1}) and after 3 h of recovery (3 h rec). Representative Western blot images are also shown. $^{*}P < 0.05$ significantly different in carriers *versus* non-carriers; $^{a}P < 0.05$ significantly different from rest; and $^{b}P < 0.05$ significantly different from postexercise.

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not influence phosphorylation of ACC (Fig. 3*A*), GSK3β (Fig. 5*B*) or protein expression of GS (Fig. 1*B*) and Akt (Fig. 4*A*). Similarly, no changes in protein expression of AS160 or phosphorylation of AktThr^{308} were noted with exercise or in the recovery period (data not shown).

Discussion

The major finding of this study was that the carriers of the *PRKAG3* mutation had an increased exercise-mediated Akt-dependent signalling response in skeletal muscle and an increased phosphorylation of AS160 compared with the non-carriers. This suggests that the mutation affects the ability to phosphorylate Akt on Ser^{473}. An increase in pAkt due to exercise is in agreement with other studies both in humans performing bicycle exercise (Sakamoto *et al.* 2004) and in rats performing treadmill exercise (Sakamoto *et al.* 2003). The finding that Akt phosphorylation on Thr^{308} was unchanged after exercise is also in agreement with a study on humans (Coffey *et al.* 2006). In an earlier study in humans, phosphorylated Akt was elevated more after resistance exercise when

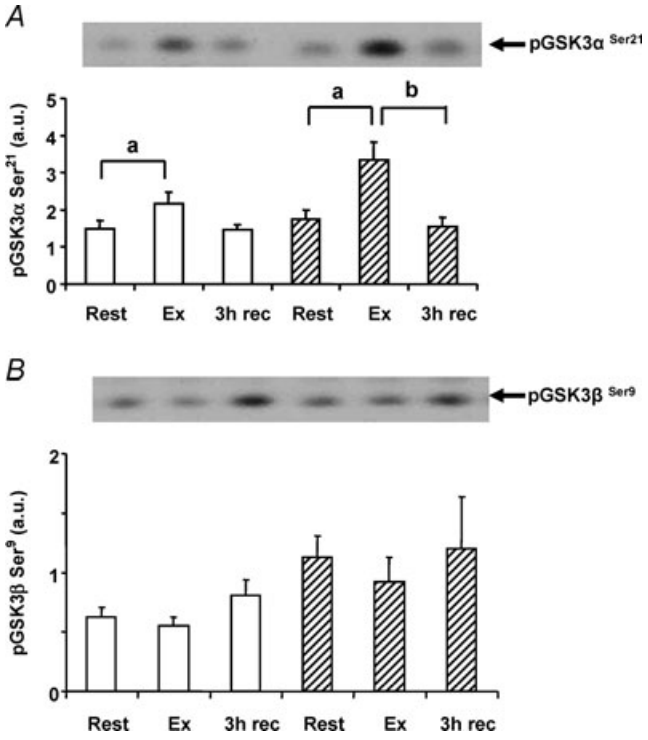


Figure 5. Effect of acute exercise on phosphorylation of GSK3α and GSK3β in skeletal muscle in non-carriers (open bars, $n = 7$) and carriers of the *PRKAG3* mutation (hatched bars, $n = 8$)
Mean values of phosphorylation of GSK3α (*A*) and GSK3β (*B*) are shown from biopsies taken from biceps femoris at rest (Rest), immediately after exercise (Ex; 700 m, 1.5 m s^{-1}) and after 3 h of recovery (3 h rec). Representative Western blot images are also shown. $^{a}P < 0.05$ significantly different from rest; and $^{b}P < 0.05$ significantly different from postexercise.

subjects had been on a high-carbohydrate diet, resulting in high muscle glycogen levels, in comparison to a low-carbohydrate diet, resulting in low muscle glycogen levels (Creer *et al.* 2005). One factor that could contribute to the higher degree of pAkt in the carriers may thus be the glycogen availability, because this differed markedly among genotypes. This study also confirms earlier studies showing that the carriers of the *PRKAG3* mutation have higher glycogen content in the skeletal muscle (Monin *et al.* 1992; Enfält *et al.* 1997). In contrast to insulin-stimulated Akt regulation, the role of Akt activation in contraction-mediated glucose uptake has been questioned, since some studies have failed to show increases in activity (Brozinick & Birnbaum, 1998; Lund *et al.* 1998; Widegren *et al.* 1998). Differences in Akt signalling in exercise may be related to the type of contraction and the intensity and duration of exercise. Of interest is that AS160 is suggested to be a site of signal integration for insulin- and contraction-stimulated signalling to GLUT4 translocation, and AS160 is also phosphorylated in rat skeletal muscle (Bruss *et al.* 2005) and in human skeletal muscle (Deshmukh *et al.* 2006) in response to contraction concomitant to Akt phosphorylation. The enhanced phosphorylation of Akt and of AS160 seen in carriers of the *PRKAG3* mutation after the type of exercise used in the present study supports an exercise-stimulated signalling, which results in an increased glucose influx into the skeletal muscle cell.

Several studies have shown that phosphorylation of Akt leads to activation of GSK3, which then influences GS activity and glycogen synthesis (Cross *et al.* 1995; Bernard *et al.* 2005). In the present study, there was also an increase in phosphorylation of GSK3α in both the carriers and the non-carriers after exercise, apart from the increased phoshorylation of Akt. No increase was seen in GSK3β, which supports an earlier study by Sakamoto *et al.* (2006) showing that pAkt is required for normal GSK3α phosphorylation but not for GSK3β phosphorylation. The GS activity in the present study did not seem to be affected by the mutation or the exercise protocol. The GS activity at rest has previously been investigated in pigs carrying the *PRKAG3* mutation by Estrade *et al.* (1994). In agreement with this study, no significant differences between the genotypes were seen despite a tendency to higher activity in the carriers. Data from the present study indicate that the rate of glycogen synthesis may be more related to glucose influx to the muscle cell. The activity of HK was significantly higher in the carriers, indicating a higher capacity for phosphorylation of glucose in the glycogen synthesis pathway. The suggestion that this pathway is affected by the mutation is also supported by the finding of upregulation of the expression and activity of the UDP-glucose pyrophosphorylase in pigs carrying the *PRKAG3* mutation (Hedegaard *et al.* 2004) and that carriers of the mutation have a higher activity of the branching enzyme (Estrade *et al.* 1994). Results from a study on transgenic

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mice (R225Q mutant $\gamma 3$) show that the mutants had an unaltered protein expression of glycogen synthase, a decreased G-6-P-independent glycogen synthase activity and an increased glycogen phosphorylase activity in skeletal muscle, which supports the results of the present study (Yu *et al.* 2006).

Interestingly, the activities of phosphorylase and the debranching enzyme of importance for glycogenolysis have previously been shown not to differ between genotypes (Estrade *et al.* 1994). In the present study, phosphorylase activity was somewhat higher in the carriers, and glycogen breakdown due to exercise did not differ between genotypes, supporting the suggestion that the mutation has no influence on glycogenolysis.

The faster resynthesis of glycogen seen in the carriers after exercise is in agreement with an earlier treadmill exercise study (Essén-Gustavsson *et al.* 2005). It has also been reported that glycogen resynthesis 2.5 h after swimming is accelerated in transgenic mice (Tg-Prkag3^{225Q}) expressing the natural pig mutation (Barnes *et al.* 2004). This study on mice suggested that the mutation caused a greater reliance on lipid oxidation and that the triglyceride content in muscle decreased in response to exercise. A study in humans with the $\gamma 3$ R225W mutation also provided evidence to suggest that the *PRKAG3* mutation influences not only glycogen metabolism but also lipid metabolism. The $\gamma 3$ R225W mutation in humans showed a 90% increase in skeletal glycogen content and a 30% decrease in intramuscular triglyceride stores (Costford *et al.* 2007). An increased lipid oxidation may reduce the demand for glucose oxidation, permitting more glucose within the muscle to be directed towards glycogen synthesis.

The *PRKAG3* mutation in pigs has been suggested to be a gain-of-function mutation, leading to an increased glucose transport and glycogen synthesis (Milan *et al.* 2000).

Interestingly, the carriers in the present study had a tendency for less total AMPK expression than the non-carriers. Phosphorylation of AMPK did not differ between genotypes in spite of the different expression of AMPK. This may indicate that AMPK might be more susceptible to phosphorylation in carriers of the *PRKAG3* mutation. It has been shown that AMPK activity in resting muscle extracts was about three times higher in non-carriers than in carriers of the *PRKAG3* mutation, both in the presence and in the absence of AMP (Milan *et al.* 2000).

In humans, exercise (Fujii *et al.* 2000; Stephens *et al.* 2002; Bruss *et al.* 2005) increases skeletal muscle AMPK activity, and ACC phosphorylation is said to be a good marker of AMPK activity (Hardie *et al.* 1998; Chen *et al.* 2000). In the study on transgenic mice (Tg-Prkag3^{225Q}) expressing the pig mutation, it was shown that after swimming the Tg-Prkag3^{225Q} mice had an increased phosphorylation of ACC compared with the wild-type control animals, but this was not seen in the recovery phase (Barnes *et al.* 2005). No changes in phosphorylation of ACC were seen in the carriers after exercise in the present study, which could be related to factors such as species difference, different fibre type composition and different exercise intensity.

In earlier studies, the CS and HAD activities have been shown to be increased and LDH decreased in a 'white' muscle, longissimus, of pigs carrying the *PRKAG3* mutation (Estrade *et al.* 1994; Lebret *et al.* 1999), while in a 'red' muscle, semispinalis capitis, the mutation had no influence on enzyme activities (Lebret *et al.* 1999). The observation in the present study that there were no differences in CS, HAD and LDH activities may be explained by the fact that younger pigs were studied and also that a different muscle, biceps femoris, was studied. An earlier study on carriers and non-carriers of the *PRKAG3* mutation has shown that biceps femoris contains 23% type I and 77% type II fibres, with no differences seen between genotypes (Essén-Gustavsson *et al.* 2005). The study mentioned earlier on longissimus, composed of 10% type I and 90% type II fibres, also showed fibre type composition to be similar between genotypes, but muscle oxidative capacity was higher in the carriers. Type II fibres primarily express AMP-kinase $\gamma 3$ (Mahlapuu *et al.* 2004) and thus metabolism in muscles containing a high proportion of type II fibres may be influenced to a greater extent by the mutation.

Glycogen in pig muscle can be separated into different pools, proglycogen and macroglycogen, and carriers have a larger pool of macroglycogen, most of which is likely to be stored in type II fibres (Essén-Gustavsson *et al.* 2005). One could speculate that the localization of glycogen granules in the cell and how they are structured together with different proteins may differ between carriers and non-carriers of the *PRKAG3* mutation and that this is an important factor to consider for the signalling response and increased glycogen synthesis.

In conclusion, increased glycogen synthesis in pigs carrying the *PRKAG3* mutation is related to an increased signalling response of Akt and its substrate, AS160, postexercise and to a higher capacity for phosphorylation of glucose.

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